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DApp Hybrid TEE-Rollup DA

1

1. 200062

DApp Hybrid TEE-Rollup

DA DA DA Python EVM gas Sepolia 406 DA 96.33% TEE DA

Hybrid TEE-Rollup DApp

: A TP311

Recoverable Challenge Protocol and Lightweight DA Cost Evaluation for Hybrid TEE-Rollup in High-Frequency DApps

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Abstract: High-frequency decentralized applications (DApps) require low latency, low on-chain overhead, and publicly verifiable arbitration. This paper proposes a recoverable challenge mechanism, an evidence-carrying compact commitment mechanism, and a data-availability-aware cost model for Hybrid TEE-Rollup. The recoverable challenge mechanism restores timeout sessions to replayable and settleable states; the compact commitment binds the state root, output hash, proof hash, DA pointer, and DA root; and the cost model analyzes payload size, batching amortization, and DA path selection. Based on a Python prototype, local EVM gas measurements, and Sepolia deployment records, the compact commitment averages about 406 bytes, the modular DA sampling path reduces amortized cost by about 96.33% under the given configuration, and the recovery path improves challenge completion under timeout scenarios. The conclusions are based on a simulated TEE, a verifiable DA registry, and a configurable cost model.

Key words: Hybrid TEE-Rollup; interactive challenge; data availability; high-frequency DApp; cost amortization; fault recovery

1 □□

AI

TEE

Hybrid TEE-Rollup TEE [1]

DApp

TEE DA EVM gas Sepolia

1

2 DA DA

3 DA DA

DA

2 □□□□

TEEROLLUP TEE Rollup

opML optimistic fraud proof

FPVM

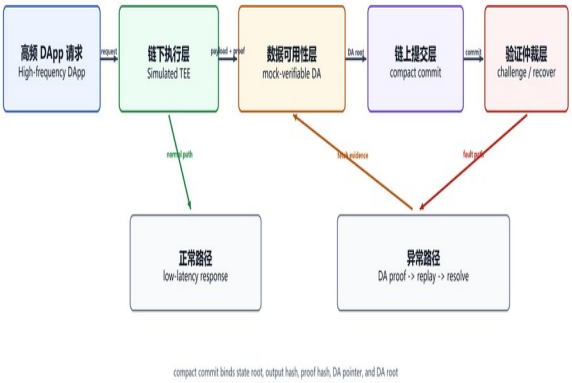
ML Hybrid TEE-Rollup

Dynamic Fraud Proof

LazyLedger DA
DA
Rollup [2]
ZK-Rollup
Optimistic Rollup
[3]
Rollup
[4][5]
[6]
Rollup [7][8][9]
[10]
Hybrid TEE-Rollup
DA
Merkle
TEE

3 □□□□

DA DA
AI



1 Hybrid TEE-Rollup
Fig.1 Prototype architecture of Hybrid TEE-Rollup
1 DApp

3.1

DApp
nonce [11]
“ _ _ ”

3.2

DA
Merkle Merkle

Merkle DA Merkle

3.3

[4][5]
DA

Solidity MockTEEVerifier
MockDARegistry MerkleVerifier HybridTEERollup
EVM Python

4 □□□□□□□□□

4.1

4.1.1

1

$$cs = (id,\ s,\ F,\ T)$$

id s F T

2

$$M = (S,\ E,\ \delta)$$

S E $\delta: S \times E \to S$

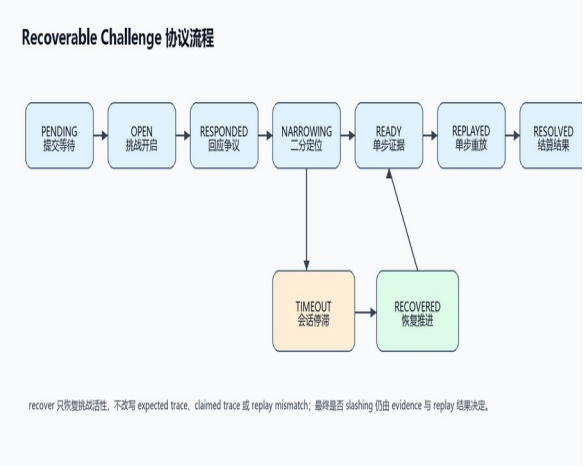
3

$$Recover(cs) = cs'$$

$$F(cs) = F(Recover(cs))$$

TIMED_OUT

4.1.2



2

Fig.2 Workflow of the recoverable challenge protocol

n log2(n)

“ ”

4.1.3

DA Merkle DA
Merkle
Merkle DA DA

“ ” “ ”

n
log2(n) 4 8 16 32 64 128 2 3
4 5 6 7

4.1.4

“ ”

“ ”

- State Root
- DA Merkle Root DA
- Payload Hash DA
- Execution Trace

-
- DA

cs Facts(cs)

$$Facts(cs_{before}) = Facts(cs_{after})$$

—— ——

DA Merkle DA

—— DA

“ ” “ ”

4.2

Merkle

4.3

DA DA EIP-4844
DA [12]

$$C_{total} = C_{commit} + C_{DA} + p_{challenge} \times C_{challenge}$$

C_commit C_DA DA

C_challenge p_challenge

$$C_{amortized} = \frac{C_{total}}{batch_{size}}$$

5 □□□□

Python TEE
DA EVM gas Sepolia RQ1
RQ2 DA
RQ3 RQ4
RQ5 Solidity EVM Sepolia

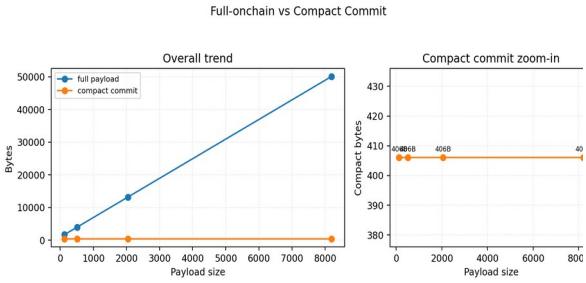
- $C_{total} = C_{commit} + C_{DA} + p_{challenge} \times C_{challenge}$ 1
- $C_{amortized} = \frac{C_{total}}{batch_{size}}$ 2

C_commit C_DA DA
C_challenge p_challenge

5.1

RQ1 Python TEE
JSON

406
8192 64 256
50068.84 50262.84 406.00



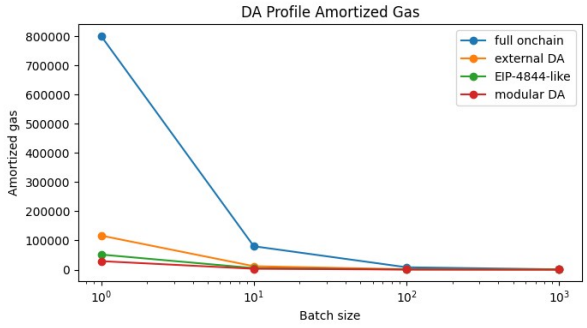
3

Fig.3 Byte-size comparison between full payloads and compact commits

5.2 DA

RQ2
DA EIP-4844 DA gas

8192
gas 1 801101.44 1000
801.21 DA 29026.98 29.03 96.33%
DA



4 gas

Fig.4 Amortized gas trends under different DA configurations

5.3

RQ3 4.1.4 Python
4 128
2

100% 0%
100%
5 EVM challengeStep gas 93807
challengeRecover gas 156891
1.67

EVM

5.4

RQ4

0%

4

Table 4 Failure scenarios and evidence boundaries

	, ,	100% 0%	
+	,	100%	
	vs	0% 100%	

5.5 Solidity EVM Sepolia

RQ5 Solidity
Hardhat EVM gas 5
Sepolia Chain ID 11155111 10825904

5 EVM gasUsed

Table 5 Average gasUsed of local EVM key paths

	gasUsed	
registerDA	113526	MockDARegistry
submitRollup	299033.67	
challengeOpen	283769	
challengeRespond	65886	
challengeStep	93807	
challengeRecover	156891	
challengeReplay	135632	
challengeResolve	113402	
challengeFinalize	33477	

Python

Hardhat

5.6

RQ1 RQ5

406 DA

6 □□

Hybrid

TEE-Rollup

DA DA

DA

Python EVM gas Sepolia

DA

TEE DA

TEE

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